BBO capstone project presentation template

Develop a concise presentation that captures:

1. **An overview of your BBO approach**
   - Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
   - Briefly describe your overall strategy and how your process fits together.

The goal of this project is to find the maximum of eight unknown functions, with 2 to 8 inputs each, using only one query per function per week. There are no formulas or gradients, only the results of my own queries, like tuning a model or running a lab experiment where every test is expensive.

The strategy: a loop that runs every week.
1. Model what I've seen. A Gaussian process (GP) learns each function from all its observations so far, and says how unsure it is everywhere else.
2. Choose the next query. I pick the point with the highest expected improvement over my best result, searching only in a "trust region" around that best point. The region grows when a query improves the result and shrinks when it doesn't.
3. Submit, learn, repeat. Each result goes back into the data for the next round.

The approach taken is tailored based on the initial description. Function 1, a radiation detector, uses a physical model of the source instead of a GP, and functions 3 and 5 are modelled on log scales because of their observed output ranges. Function 7 turned out to be a known benchmark (Hartmann-6), but I treat it as a black box, because using the answer would defeat the purpose of the project.

2. How your strategy has evolved

- Describe the key changes in your approach since your early rounds.
- What influenced these changes – data trends, feedback, model performance or intuition?
- Identify which principles or heuristics now guide your query decisions.

Key changes

" Weeks 1 – 2: a GP with an upper-confidence-bound rule, tuned by hand each week, plus judgement from plots.

" Week 3: normalised outputs, one length scale per input, and gradient-based search; in 8D, my grid had only tested 3 values per input.

" Weeks 4 – 8: BoTorch, with hand-tuned length-scale bounds and exploration settings for each function.

" Week 9: a rewrite based on tests: default priors, Log Expected Improvement, a search box around the best point, log scales, and a source model for function 1.

" Week 10+: the search box adapts each week (TuRBO).

What influenced them

" Data trends: most best values came in weeks 3 – 5, and later, highly exploratory queries landed on edges and corners and rarely helped.

" Model performance: 30 of 34 hand-set length scales sat on their bounds, so the bounds were deciding the model, and the acquisition optimiser silently returned poor points for two functions.

" Evidence and the brief: leave-one-out tests and simulations of the final rounds decided between options, and the function descriptions told me about ceilings and peaks.

Principles that guide me now

" Let the data set the model, not hand-tuning.

" Exploit near the best region, where the model is most reliable, and take risks only where a miss costs nothing.

" Make every decision a rule in code, so the same data always gives the same query.

3. **Patterns, data and insights**
   - Highlight the most meaningful trends you've observed across your data points or model evaluations.
   - Identify which variables or behaviours seem to influence results the most.
   - Explain how these observations have shaped your understanding of the search process.

" Diminishing returns. Seven of eight functions beat their initial data, mostly by week 5 (function 5 went from 1,089 to 8,662). Since then, the gains have been small.

" A few inputs drive most of the variation. The GP's length scales show that x1 dominates function 2, x3 function 3 and x5 function 7, while x8 barely matters in function 8.

" Good results cluster. The best points of functions 1, 4, 6, 7 and 8 sit close together; for functions 2 and 3, they form bands along a single input.

" Function 1 hides a signal in tiny numbers. Its readings are almost all zero, but their size falls off smoothly with distance from one point, which locates a radiation source.

What this taught me: with so few points, the model is reliable only near data I've already collected, so the search became local. The descriptions in the brief turned out to be as valuable as the data.

4. Decision-making and iteration

- Explain how you balance exploration and exploitation.
- Describe one or two examples of strategic decisions. What worked, what didn't, and why?
- Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

Balancing exploration and exploitation: expected improvement weighs a high prediction against high uncertainty without a hand-tuned knob. The trust region keeps the search near the best point, doubling after an improvement and halving after a miss. With few rounds left, I lean towards exploitation.

Strategic decisions

" What worked: replacing hand-tuning with the GP's default priors and expected improvement.

" What didn't: that exploratory setting, in weeks 6 – 8, pushed queries to edges and corners and produced some of my worst values.

" A deliberate bet: for function 1, I query the predicted centre of the source, even though the strongest reading nearby was negative. A miss costs nothing, since its best is effectively zero, and a hit could jump to about 0.76.

Handling uncertainty. I test before trusting: leave-one-out predictions, checks that a query doesn't hinge on a single point, and simulations. I write down assumptions and fallbacks, such as function 1's weaker second source if the bet fails, and the trust region adapts on its own when results don't match expectations.

5. **Next steps and reflection**
   - Outline your planned actions or experiments to improve the model's performance.
   - Include two or three sentences connecting your work to the broader ML landscape.
   - Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

Planned actions

" Compare each prediction with its actual result as new weeks arrive, to find where the GP is over-confident, starting with function 7, where it ranks unseen points worst.

" If function 7's GP keeps over-predicting, combine it with a support vector regressor, which predicted that function better in my tests.

" If function 1's source reads negative, target the weaker second source.

" In the final round, exploit only: there's no later round to benefit from exploring.

The broader ML landscape. This loop of model, uncertainty, decision and query is how hyperparameter-tuning tools such as BoTorch, Ax and Optuna work, and how automated labs choose experiments in drug discovery and materials science. The lessons carry over: when every evaluation is expensive, calibrated uncertainty and domain knowledge matter more than model complexity, and documenting data and models, with datasheets and model cards, keeps results reproducible.

For a non-technical audience. "Imagine finding the best recipe with one taste test a week. I keep a map of what I've learned, test the most promising spot near the best recipe so far, and update the map each week. Seven of eight recipes have improved, and the last one is a needle in a haystack that I have a plan for."